51951 MP-295

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VI Semester III Year B.C.A. Examination, May/June -2025

(Scheme : NEP)

COMPUTER APPLICATION

Artificial Intelligence and Applications

Time : 2½ Hours Max. Marks : 60

Instruction : Answer part - A and Part - B.

PART - A

Answer all the questions : [5 × 10 = 50]

1) a) Explain the key disciplines that contributed to the foundation of AI. [5]

b) Describe the working of a simple reflex agent. [5]

OR

c) Differentiate the following agents : [6]

i) Fully observable and partially observable

ii) Dynamic and static

iii) Single and Multi Agent.

d) Explain Goal Base agent [4]

2) a) Describe the components of a well defined problem. [4]

b) With an example, Describe the working of Breadth - first search. [6]

OR

c) Write the structured formulation for 8 puzzle problem [4]

d) Briefly explain A* search algorithm. [6]

P.T.O.

3) a) In detail, describe the working principle of wumpus world environment. [10]

OR

b) What is forward chaining? How does it work? Explain. [5]

c) What is first order logic? Specify the syntax of first order logic in BNF form. [5]

4) a) List and define main types of machine learning. [5]

b) With an Example, Explain decision tree. [5]

OR

c) Differentiate regression and classification. [4]

d) Write a note on artificial neural network. [6]

5) a) Explain the natural language tasks. [4]

b) Briefly explain Essential steps required for image processing. [6]

OR

c) Write a note on computer vision. [5]

d) What is Robotics? List the applications of robot and mention its role. [5]

PART - B

Answer any five questions : [5 × 2 = 10]

6) What is an Agent?

7) What is PEAS?

8) Differentiate Informed and uninformed search

9) What is an heuristic function?

10) Differentiate propositional logic and first order logic.

11) Define modus ponens.

12) What are the advantages of support vector machine?

13) What is Information Retrieval?